DOCKER THEORY

DOCKER:

Docker :-

Container orchestration

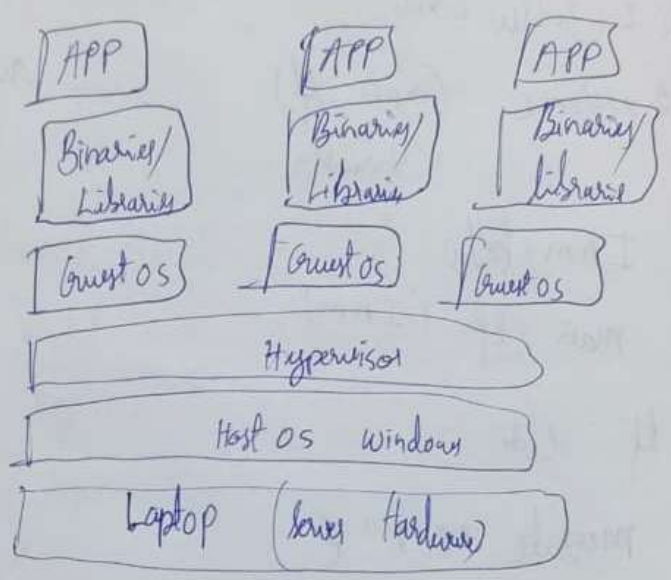
ephemeral in nature

Docker is an open platform for developing, shipping and running application. Docker enables you to seperate your application from your infrastructure so you can deliver software quickly.

with Docker you can manage your infrastructure in the same ways you manage your application.

VIRTUALIZATION:

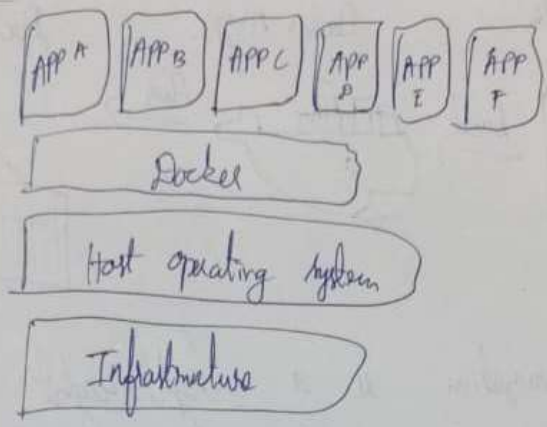
Virtualization :-



Virtualization is a technology that creates virtual, simulated versions of computing resources like servers, storage, networks or operating systems from a single physical machine allowing multiple isolated virtual environments (Virtual machines VM's) to run simultaneously. Sharing hardware resources efficiently.

CONTAINERIZATION:

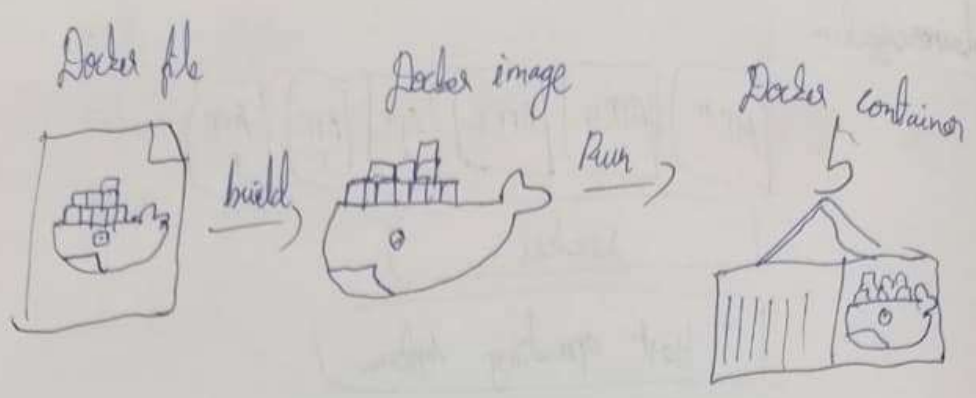
Containerization :



lightweight software containers share the machine OS system kernel and therefore do not require an OS per application, driving higher server efficiencies and reducing server and licensing cost.

Containerization :

Containerization in docker is a lightweight form of operating system level virtualization that packages an application and all its dependencies into a single standardized isolated unit called a container. Docker provide the tools platform for building sharing and running these containers consistently across any environment from a developer local machine to a production cloud server.



Containerization is a lightweight virtualization that involves packaging an application and its dependencies into a 'container' unlike traditional Virtual machine (VM's) which include an entire operating system (OS), containers share the host OS kernel and isolate the application processes from the rest of the system.

KEY COMPONENTS OF DOCKER:

Key components of Docker

Docker engine : The runtime that creates and manages containers

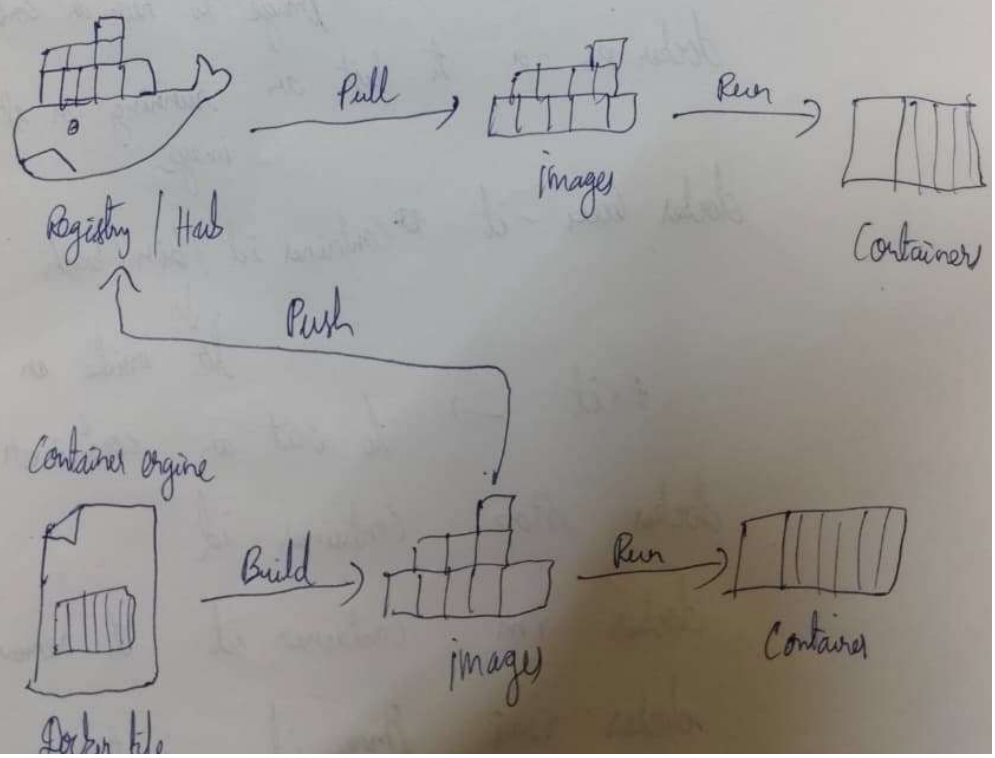
Docker images : Read only templates that defines the contents of a container.

Docker containers : Instances of Docker images running as isolated processes.

Docker file : A script that contains instructions for building a docker image

Docker ~~images~~ Hub :- A public registry for sharing and managing docker images.

DOCKER REGISTRY:

Docker registry :-

A docker registry is a centralized storage and distribution system for docker images. which are the building blocks for running application in containers. it function like library or a package manager, allowing user to upload (push) and download (pull) container images efficiently.